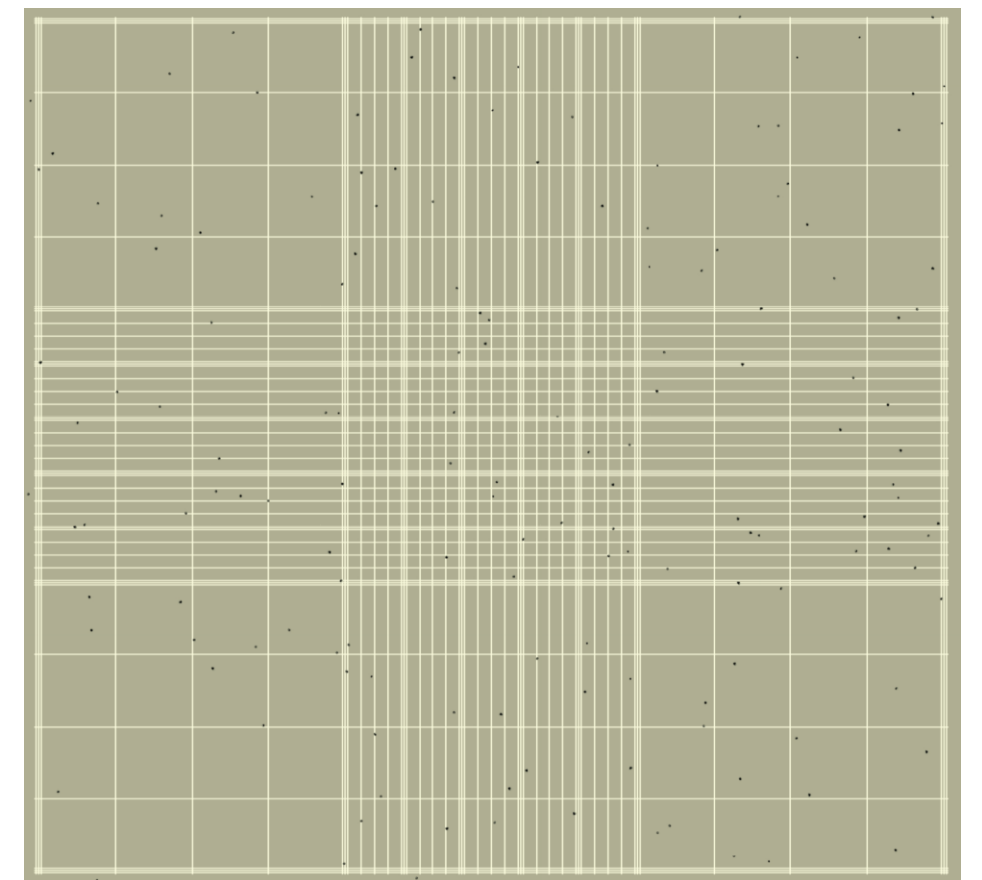


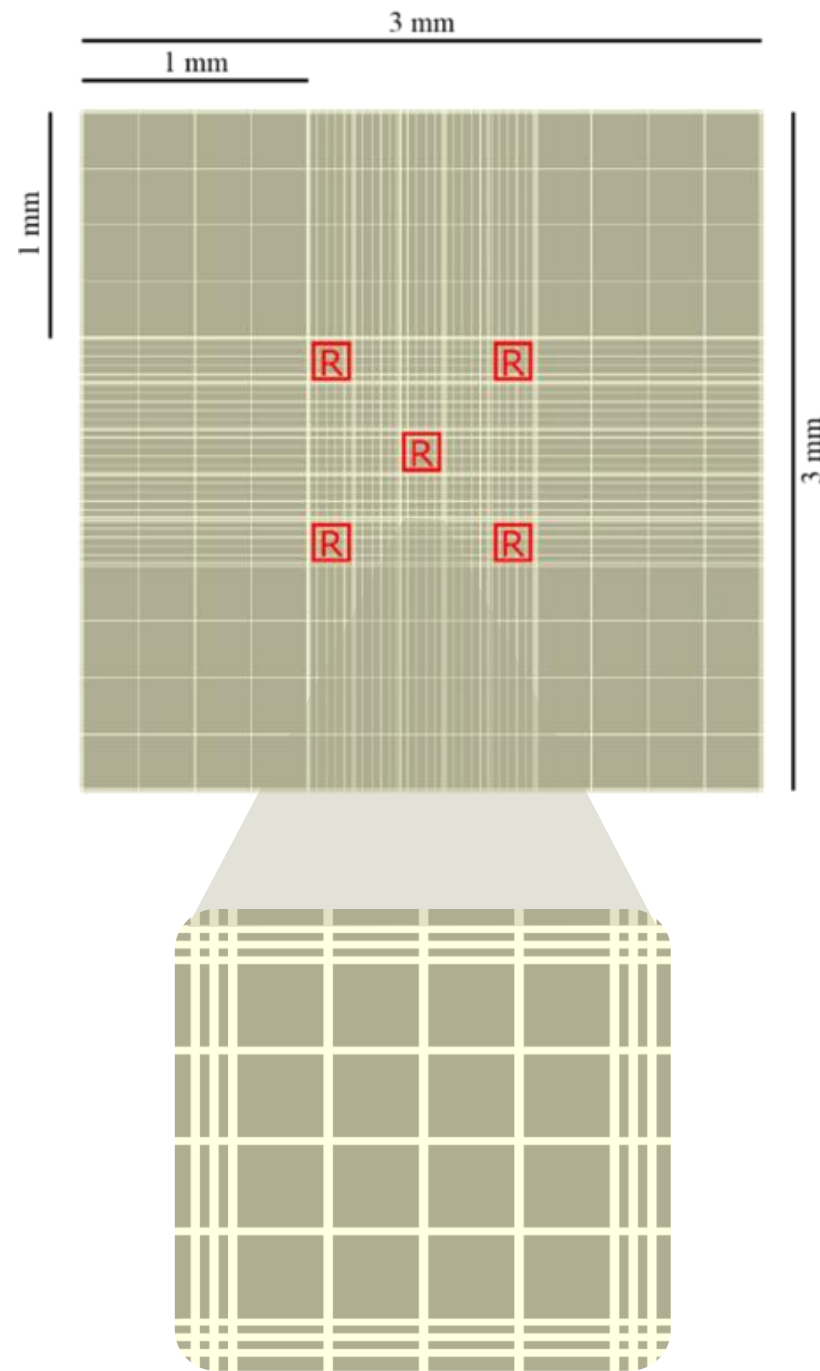
Manual Cell Counts

Background & Indications

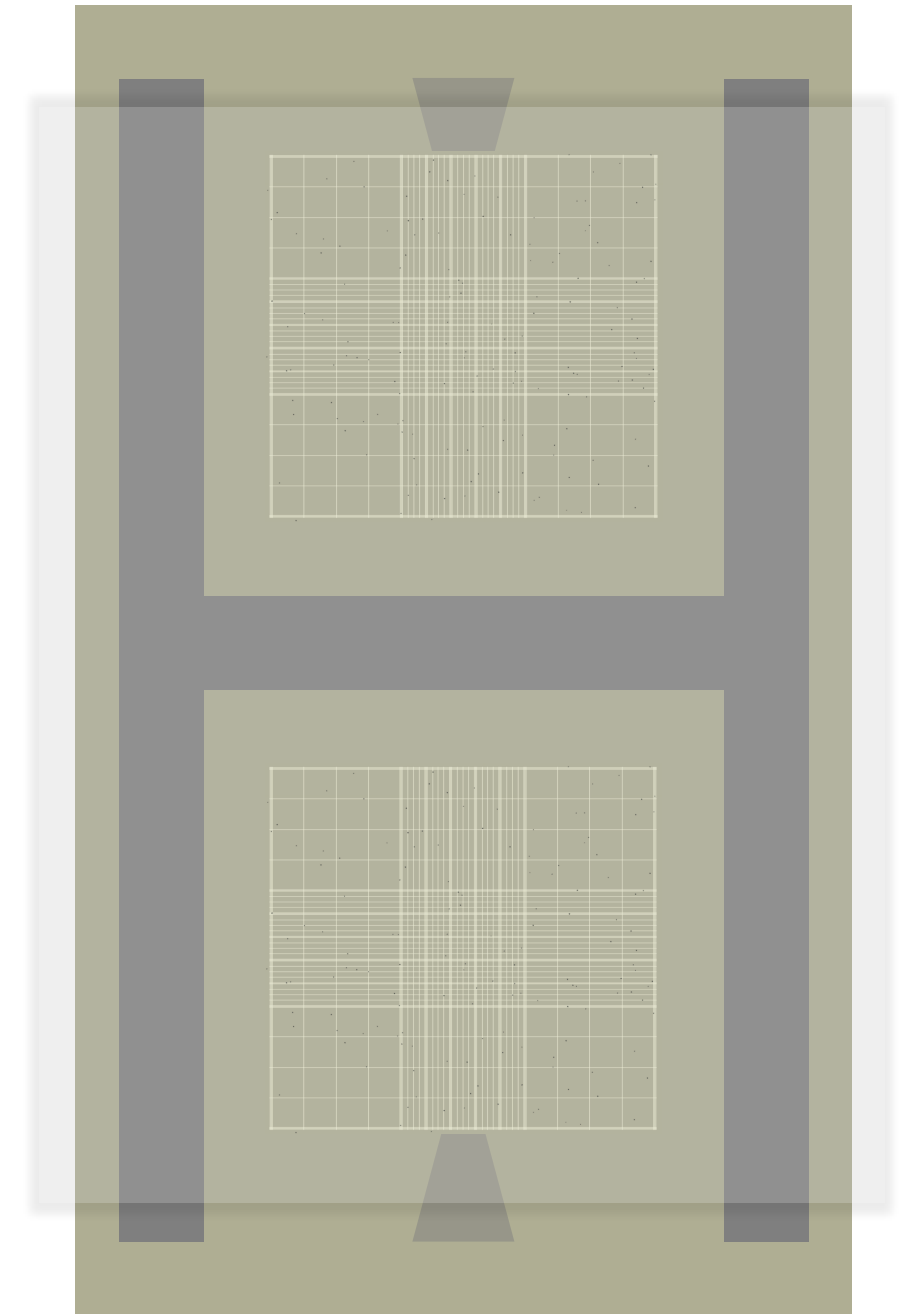
- **Primary Method:** Automated CBCs
- **Indications for Manual Counts:**
 - Instrument failure or breakdown
 - Erroneous automated counts
 - Automated counts below instrument linearity
 - Verification of slide estimates to instrumentation mismatches
 - Analysis of body fluids
- **Key Takeaway:** Manual counts are the gold-standard backup



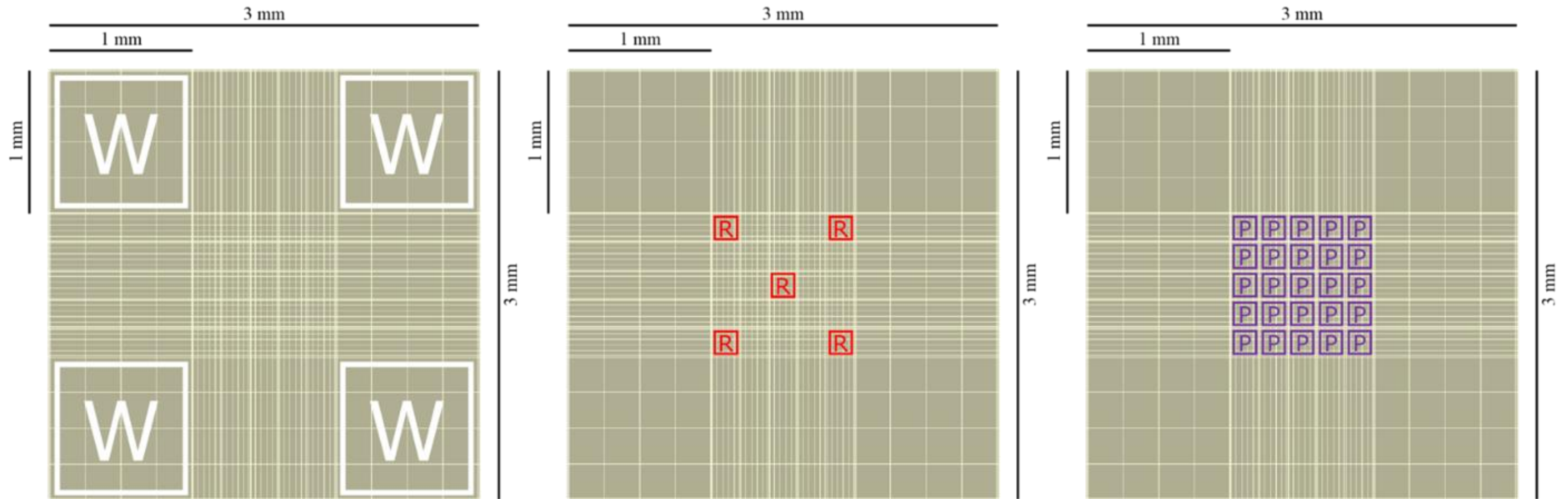
Neubauer Hemocytometer



- **Chambers:** Two identical ruled counting
- **Cover Glass:** Maintains a fixed chamber depth of 0.1 mm
- **Overall Ruled Grid (per side):**
 - Total Area: $3\text{mm} \times 3\text{mm} = 9\text{mm}^2$
 - Divided into **9 large squares** ($1\text{mm} \times 1\text{mm} = 1\text{mm}^2$ each)
- **Central Large Square (1mm^2)**
 - Subdivided into **25 small squares** ($0.2\text{mm} \times 0.2\text{mm} = 0.04^2$ each)
 - Each small square is bordered by triple lines

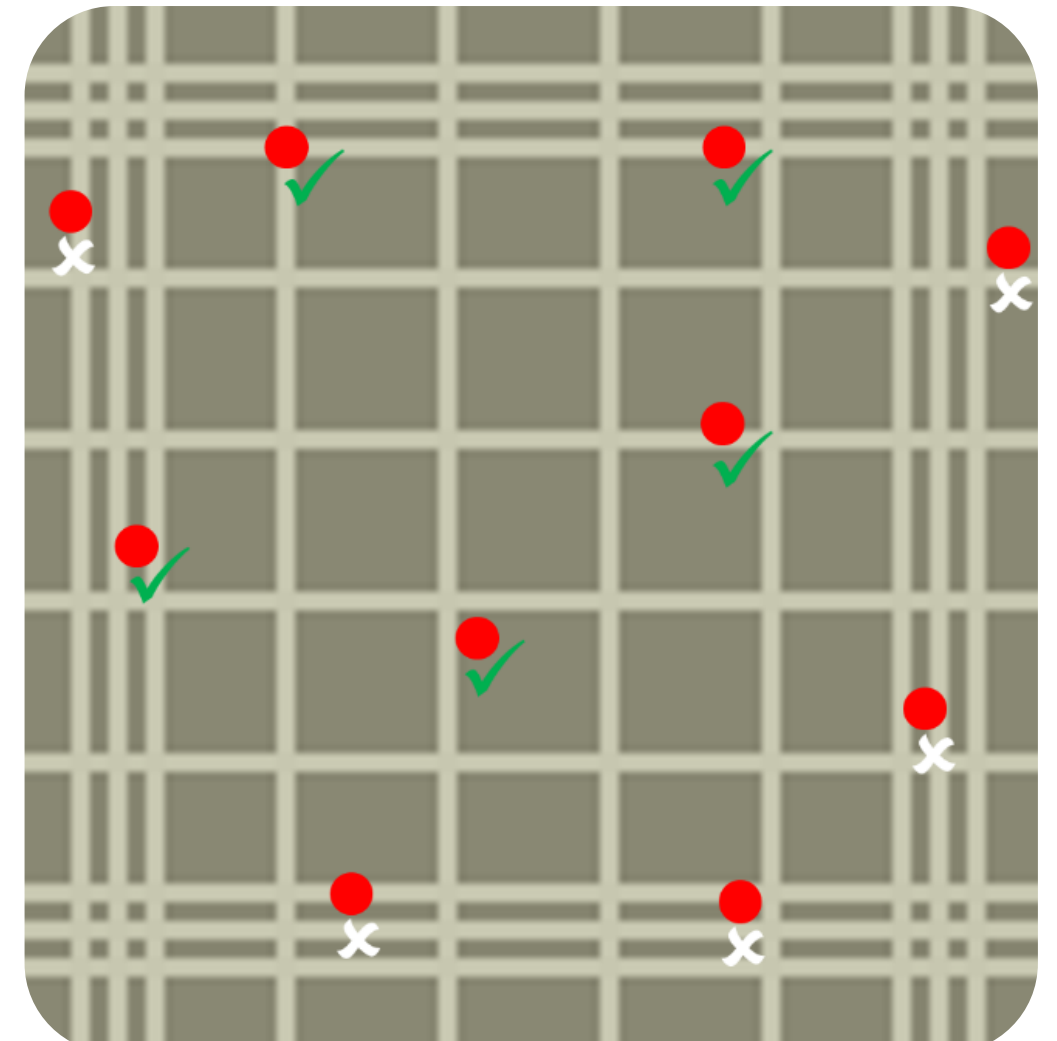


Counting Areas by Cell Type



Boundary Counting

- **Triple-Line Ruling:** Middle line serves as the true boundary
- **Inclusion Rule:** Count cells completely inside the square OR touching the top or left middle boundary lines
- **Exclusion Rule:** Omit cells touching the bottom or right middle boundary lines
- **Purpose:**
 - Prevents double-counting cells along shared borders
 - Compensates for random marginal cell distribution



Cell Concentration Formula

$$\text{Cells}/\mu\text{L} = [\text{Average cells counted}] * [\frac{1}{\text{Area}}] * [\frac{1}{\text{Depth}}] * [\text{Dilution factor}]$$

- **Average Cells Counted:**
 - (Side 1 Count + Side 2 Count) / 2
- **Depth:** Fixed at 0.1mm
- **Area Counted:**
 - Sum of area of squares counted on ONE side
- **Dilution Factor:**
 - (Diluent Volume + Sample Volume) / (Sample Volume)

Calculation Worked Example

Problem: 68 cells counted on Side 1; 72 cells counted on Side 2 across all 9 large squares. Dilution: 20 μL blood diluted with 980 μL diluent.

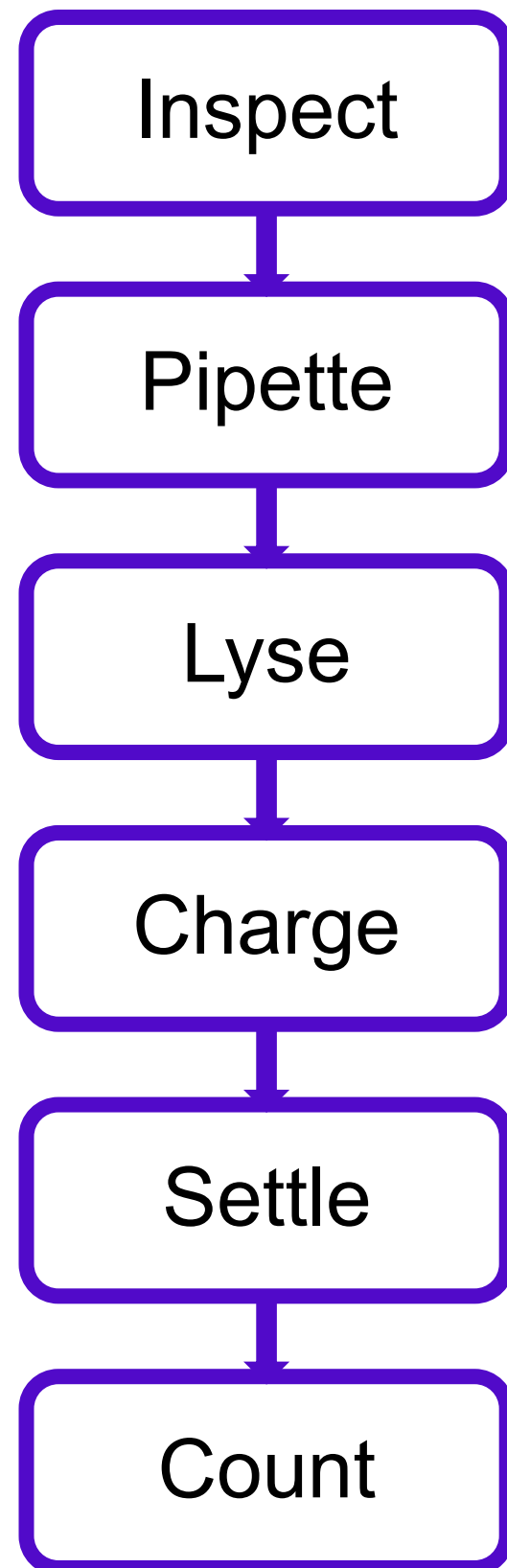
$$\begin{aligned}\text{Cells}/\mu\text{L} &= \left[\frac{68+72}{2} \right] * \left[\frac{1}{9} \right] * \left[\frac{1}{0.1} \right] * \left[\frac{20+980}{20} \right] \\ &= 3,889 \text{ WBC}/\mu\text{L} = 3.9 \times 10^3/\mu\text{L} = 3.9 \times 10^9/\text{L}\end{aligned}$$

Manual WBC via Leuko-TIC

Principle: Acetic acid diluent lyses non-nucleated RBCs while preserving WBCs and nucleated cells. Gentian violet stains WBC nuclei violet-blue



- **Specimen:** EDTA whole blood (< 4 hours post-collection)
- **Reagent Composition:**
 - **Volume:** 380 μ L diluent per vial
 - **Components:** Acetate buffer pH 3.0, Gentian violet (0.1 g/L 0.1 g/L), Detergent (< 0.3%).
 - **Dilution:** 20 μ L specimen + 380 μ L diluent
 - = 1:20 dilution (DF = 20)



Leuko-TIC Procedure

1. **Inspection:** Check EDTA specimen for clots or fibrin strands
2. **Pipetting:** Add 20 μ L well-mixed blood to Leuko-TIC vial (pre-wet tip, gently wipe excess from outside of tip)
3. **Lysis:** Mix vigorously and wait minimum 5 minutes for complete RBC lysis
4. **Charging:** Mix vial again; use 10 μ L to charge each hemocytometer side (do not overfill/underfill)
5. **Settling:** Place hemocytometer in a humidified petri dish for 5 minutes
6. **Counting:** Under 40x objective, count WBCs in 4 large corner squares on Side 1, then repeat on Side 2

WBC Reference Intervals

Population	Conventional Units ($\times 10^3 / \mu\text{L}$)	SI Units ($\times 10^9 / \text{L}$)
Adults (General)	4.5 – 11.0	4.5 – 11.0
Adults (African Descent)	3.0 – 9.0	3.0 – 9.0
Children (4 years)	4.5 – 13.5	4.5 – 13.5
Infants (6 months)	6.0 – 18.0	6.0 – 18.0
Newborns (Birth)	9.0 – 30.0	9.0 – 30.0

PBS WBC Estimation

- **Magnification:** 40x or 50x high dry field (HPF)
- **Procedure:**
 1. Locate monolayer
 2. Count WBCs in 10 fields (include fields with 0 WBCs)
 3. Calculate average per HPF and multiply by 2,000
- **Interpretation:**
 - Normal range: 2 to 6 WBCs/HPF ($\approx$ 4,000–11,000 WBC/ μ L)
 - Valid for WBC counts $< 25,000/\mu$ L

nRBC Interference & Correction



- **Problem:** Acetic acid lyses non-nucleated RBCs, but nRBCs contain nuclei and are counted as WBCs on the hemocytometer, causing falsely elevated WBC counts
- **Correction Criteria:** Perform correction if > 5 nRBCs per 100 WBCs are seen on the 100-WBC differential count

$$\text{Corrected WBC Count} = \frac{\text{Uncorrected WBC}/\mu\text{L} \times 100}{100 + \text{nRBC count per 100 WBC}}$$

- **Reporting:** Report the corrected WBC value along with a comment stating count was adjusted for nRBCs

nRBC Correction Worked Example

- **Example Scenario:**
 - Chamber Uncorrected WBC: $45 \times 10^9/\text{L}$ ($45,000/\mu\text{L}$)
 - Differential Count: 50 nRBCs observed per 100 WBCs
- **Calculation:**

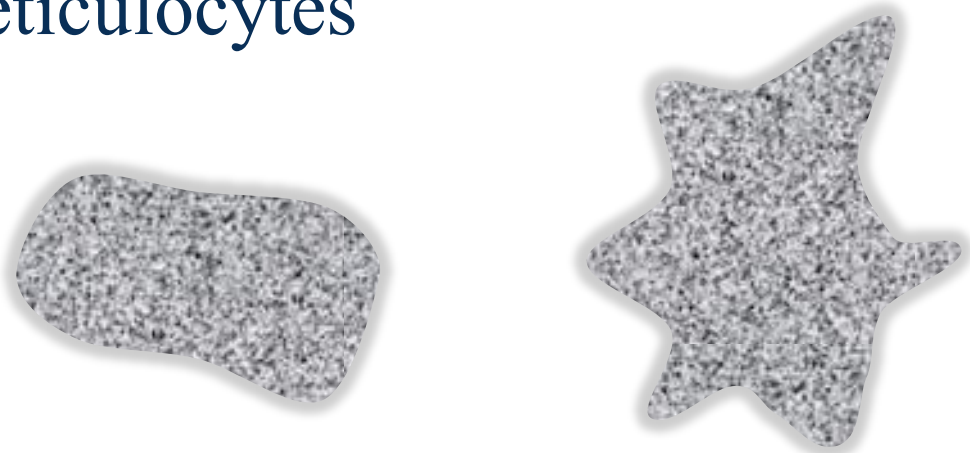
$$\text{Corrected WBC} = \frac{45,000 \times 100}{100 + 50} = \frac{4,500,000}{150} = 30,000/\mu\text{L} = 30 \times 10^3/\mu\text{L} \quad (30 \times 10^9/\text{L})$$

$$\text{Total Nucleated Count (TNC)} = 45.0 \times 10^3/\mu\text{L}$$

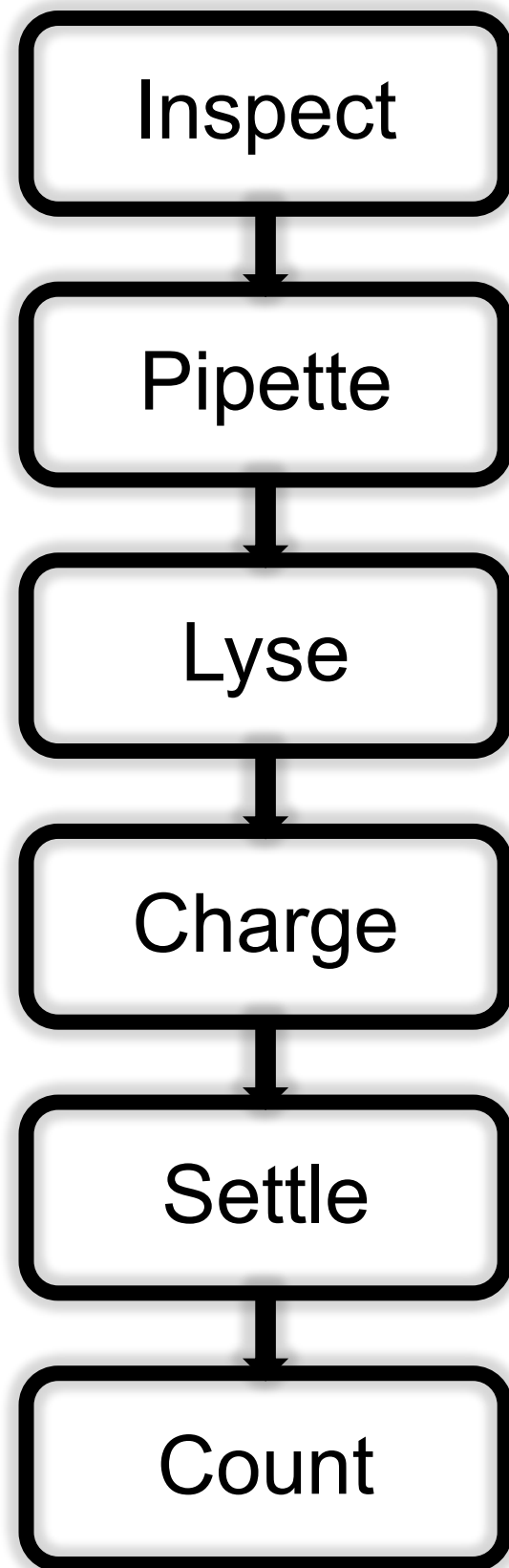
$$\text{Corrected White Blood Cell Count (WBC)} = 30.0 \times 10^3/\mu\text{L}$$

Manual PLT via Thrombo-TIC

Principle: Whole blood added to 1% ammonium oxalate diluent (pH 6.0). Lyses RBCs while preserving platelets, leukocytes, and reticulocytes



- **Appearance:** Under phase contrast, appear as refractile, oval/round structures with fine dendritic processes
- **Reagent Volume:** 990 μL diluent per vial
 - **Components:** Acetate buffer pH 3.0, Gentian violet (0.1 g/L 0.1 g/L), Detergent ($< 0.3\%$).
 - **Dilution:** 10 μL specimen + 990 μL diluent
 - = 1:100 dilution (DF = 100)



Thrombo-TIC Procedure

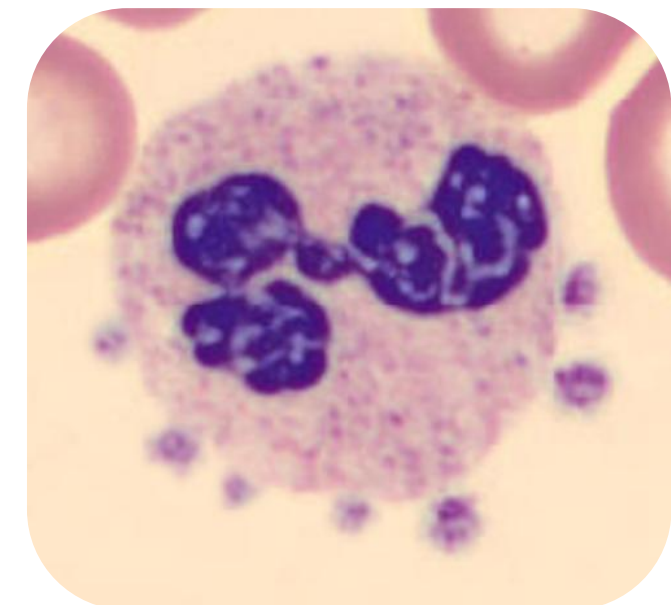
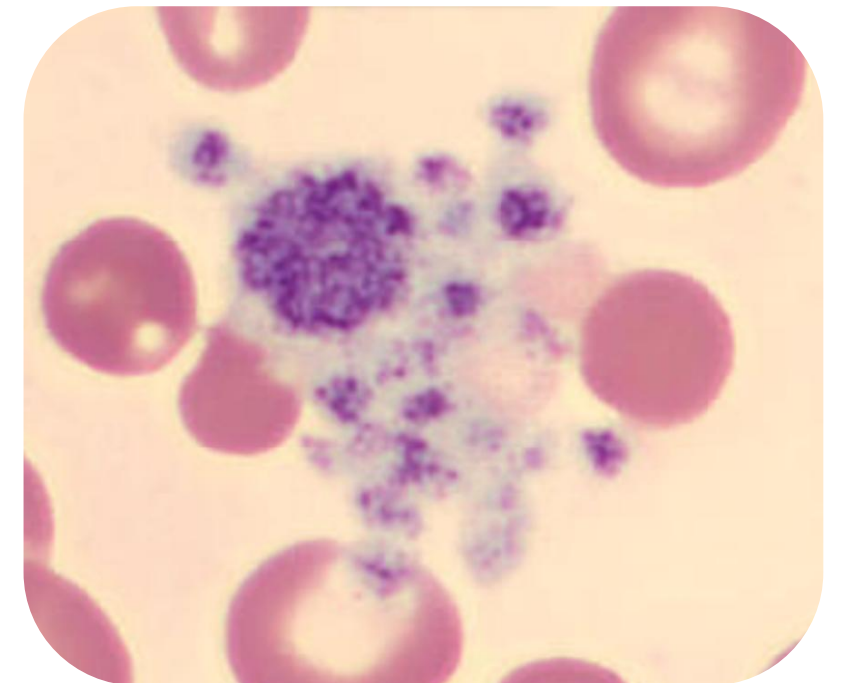
1. **Inspection:** Check EDTA specimen for clots or fibrin strands
2. **Pipetting:** Add 10 μ L well-mixed blood to Thrombo-TIC vial (pre-wet tip, gently wipe excess from outside of tip)
3. **Lysis:** Mix vigorously and wait minimum 5 minutes for complete RBC lysis
4. **Charging:** Mix vial again; use 10 μ L to charge each hemocytometer side (do not overfill/underfill)
5. **Settling:** Place hemocytometer in a humidified petri dish for 5 minutes
6. **Counting:** Under 40x objective, count PLTs in 25 small squares on Side 1, then repeat on Side 2

PLT Reference Range & Estimates

- **Normal Reference Range (Adults):**
 - $150 - 400 \times 10^3 / \mu\text{L}$ ($150 - 400 \times 10^9 / \text{L}$).
- **PBS Platelet Estimation (100x Oil Immersion):**
 - Examine 10 fields in monolayer
 - Normal: **7 to 20 platelets per field**
 - Formula: $\text{Average PLTs/field} \times 20,000 = \text{Estimated PLT}/\mu\text{L}$.
- **Low Field Scan (10x):** Always check feathered edge and margins for **platelet clumping**, giant platelets, or platelet satellitosis

Platelet Clumping & Sodium Citrate Adjustment

- **EDTA-Induced Pseudothrombocytopenia:** Some patient antibodies cause platelets to spontaneously clump in EDTA tubes
- **Action:**
 - If clumps are seen, **DO NOT** report the EDTA platelet count
 - Redraw patient in a **Sodium Citrate (Blue Top)** tube
- **Dilution Factor Correction for Sodium Citrate:**
 - Liquid citrate dilutes blood by 1: 9
 - $\text{Corrected PLT Count} = \text{Citrate Tube PLT Count} \times 1.1$



Sources of Error

- x **Specimen Sampling:** Inadequate mixing of blood tube prior to pipetting; presence of micro-clots
- x **Pipetting & Wiping:** Failure to wipe excess blood from outside of pipette tip
- x **Chamber Loading:** Overfilling or underfilling hemocytometer chambers; bubbles under coverslip
- x **Evaporation:** Allowing chamber to dry out during incubation (always use humidified petri dish)
- x **Debris Interference:** Bacteria, dust, or reagent debris mistaken for platelets (verify refractility under phase contrast)
- x **nRBC Interference:** Failing to correct WBC counts when > 5 nRBC/100 WBC are present